

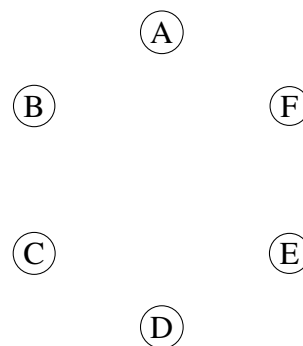
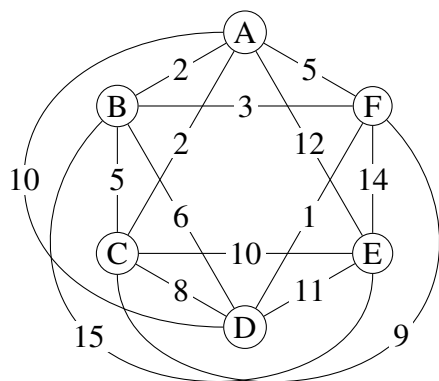
The Sorted Edges / Cheapest Link Algorithm

Steps: Add the next cheapest edge to your circuit **unless**

1. it closes the circuit too soon, or
2. creates a degree 3 vertex.

Break ties by choosing the alphabetically smallest edge.

Apply the Sorted Edges Algorithm to find a Hamiltonian circuit. Draw in the edges, labeled with their weight, as you add them on the empty graph.



Sorted edges	weight	used? (or why not)
<i>FD</i>	1	
<i>AB</i>	2	
<i>AC</i>	2	
<i>BF</i>	3	
<i>AF</i>	5	
<i>BC</i>	5	
<i>BD</i>	6	
<i>CD</i>	8	
<i>CF</i>	9	
<i>AD</i>	10	
<i>CE</i>	10	
<i>DE</i>	11	
<i>AE</i>	12	
<i>EF</i>	14	
<i>BE</i>	15	

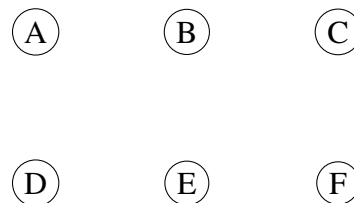
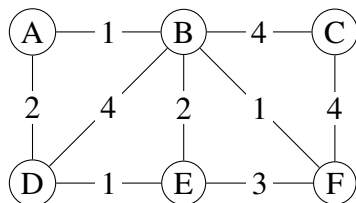
List the vertices of the Hamiltonian circuit, starting at vertex A.

Total weight of the circuit?

Do you think the circuit we obtained in the best possible?

Can you construct a graph such that the Sorted Edges Algorithm will never result in a Hamiltonian circuit of smallest weight? What does this tell us about the Sorted Edges Algorithm?

What happens if you apply Sorted Edges/Cheapest Link to the following graph?



Sorted edges	weight	used? (or why not)
<i>AB</i>	1	
<i>BF</i>	1	
<i>DE</i>	1	
<i>AD</i>	2	
<i>BE</i>	2	

Sorted edges	weight	used? (or why not)
<i>EF</i>	3	
<i>BC</i>	4	
<i>BD</i>	4	
<i>CF</i>	4	

What is the problem here?

There is a Hamiltonian circuit on this graph. What is the smallest-weight Hamiltonian circuit you can find?

Circuit: _____ Weight: _____